

# Nate Cirino

832-833-5202 | [cirino.na@northeastern.edu](mailto:cirino.na@northeastern.edu) | [portfolio](#) | [linkedin.com/in/nate-cirino](https://www.linkedin.com/in/nate-cirino) | [github.com/nateci](https://github.com/nateci)

Available May 2027

## EDUCATION

### Northeastern University

May 2027 | MGPA: 3.6/4.0

*B.S. in Computer Science, Systems Concentration*

*Boston, MA*

**Coursework:** Computer Systems, Digital Design & Computer Architecture, Systems Security, Computer Networks, Cybersecurity, Algorithms & Data Structures, Object-Oriented Design, Foundations of Data Science, Foundations of Software Engineering

## PROJECTS

### Kepler | *Rust, C++, LSM-tree, WAL, MVCC, Raft, Tokio*

Mar. 2026 – Present

- Built a distributed, linearizable key-value store in Rust/C++ with a from-scratch LSM-tree storage engine
- Implemented the on-disk storage path end to end: memtable, write-ahead log (WAL), and leveled SSTables
- Built single-tier compaction with full SSTable rewrite every 4 flushes; inline, writer-blocking execution
- Implemented Raft election, log replication, and membership changes; added MVCC layer for snapshot isolation

### NOAA AUV Sonar Detection | *Python, PyTorch, AWS Neuron SDK, SageMaker*

April 2025 – July 2025

- Designed AWS Neuron SDK deployment path for onboard custom-silicon inference under hard real-time limits
- Trained and evaluated CNN models on thermal sensor spectrograms to detect novel aquatic signatures for AUVs
- Refined architecture and loss formulation to address class imbalance and sensor noise, improving detection

### Sentinel | *Next.js, FastAPI, Claude, pgvector, OpenTelemetry, Docker*

Apr. 2026 – Present

- Built an AI agent applying LLM tool-use to autonomously triage incidents from Prometheus, Tempo, and Loki telemetry
- Built pgvector RAG over runbooks and postmortems with prompt caching on frozen schemas; 100% eval accuracy
- Human-in-the-loop remediation gated behind approval; running live in SculptAI production triaging incidents

## EXPERIENCE

### Wayfair

Jul 2026 – Present

*Software Engineering Co-op*

*Boston, MA*

- Own front-of-house device stack across all Wayfair physical retail brands (AllModern, Birch Lane, Joss & Main, Perigold): e-label devices, mobile, and customer/associate-facing POS
- Building a supplier-agnostic e-label API unifying 4 hardware suppliers behind one device interface, powering new outlet launches
- Shipped POS checkout flows: in-store scan-to-basket customer switching and live associate-entry customer sync

### Wolters Kluwer

Jul 2025 – Jun 2026

*Software Engineer*

*Boston, MA*

- Overhauled Relay JMS broker serialization with json-io across 20+ microservices, eliminating timeouts at scale
- Implemented OIDC/IdP authentication via Azure across 20 enterprise health systems, securing 10K+ clinicians
- Integrated Expert AI into CME accrual/redemption microservices, expanding GenAI CDS to 250K+ clinicians
- Built Java Spring microservice for CME credit issuance inside Expert AI's clinician-facing workflow
- Optimized Expert AI search and filtering across 20+ microservices, improving clinical query relevance at scale

### SculptAI

Jan 2025 – Aug 2026

*Software Architect*

*New York, NY*

- Led eng org restructure at \$1M-funded startup, partitioning teams into DevOps, dev, and agentic workflows; continue as technical advisor on architecture direction
- Maintained 99.99% uptime across 1-2M monthly model calls for 2K paying users via AWS WAF, Datadog, and K8s
- Architected Flask backend with RAG pipeline for personalized workout recommendations using Bedrock KB
- Evolved RAG stack from LangChain to Vespa to zero-ops Bedrock KB, improving retrieval and reducing overhead

### Chevron New Energies

May 2024 – Aug 2024

*Software Engineer Intern*

*Houston, TX*

- Built RESTful APIs and data pipelines using Azure Data Factory, improving product efficiency by 40%
- Implemented CI/CD workflows using Azure DevOps and K8s, enabling automated deployment and rollback

## SKILLS & HONORS

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**Languages/Tools:** C, C++, Rust, Python, Bash, Linux, Java, TypeScript, JavaScript, Docker, Kubernetes, AWS, Azure, Terraform, Jenkins, Buildkite, Groovy, PostgreSQL, GraphQL

**Frameworks & Libraries:** PyTorch, TensorFlow, Pandas, FastAPI, Flask, Node.js, Express, React, Next.js

**AI / Agentic Tooling:** Devin, Cursor, Claude Code, Claude Cowork, GitHub Copilot, CodeRabbit

**Honors:** Dean's List (All Semesters), Tech Lead at Northeastern Electronic Racing, PawHacks, json-io OSS Contributor